

Round Robin is optimal for fault-tolerant broadcasting on wireless networks[☆]

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Abstract

We study the completion time of broadcast operations on *static ad hoc wireless networks* in presence of *unpredictable* and *dynamical* faults.

Concerning *oblivious* fault-tolerant distributed protocols, we provide an $\Omega(Dn)$ lower bound where n is the number of nodes of the network and D is the source eccentricity in the fault-free part of the network. Rather surprisingly, this lower bound implies that the simple *Round Robin* protocol, working in $O(Dn)$ time, is an *optimal* fault-tolerant oblivious protocol. Then, we demonstrate that networks of $o(n/\log n)$ maximum in-degree admit faster oblivious protocols. Indeed, we derive an oblivious protocol having $O(D \min\{n, \Delta \log n\})$ completion time on any network of maximum in-degree Δ .

Finally, we address the question whether *adaptive* protocols can be faster than oblivious ones. We show that the answer is negative at least in the general setting: we indeed prove an $\Omega(Dn)$ lower bound when $D = \Theta(\sqrt{n})$. This clearly implies that no (*adaptive*) protocol can achieve, in general, $o(Dn)$ completion time.

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1. Introduction

Static ad hoc wireless networks (in short, wireless networks) have been the subject of several works in recent years due to their potential applications in scenarios such as battlefields, emergency disaster relief, and in any situation in which it is very difficult (or impossible) to provide the necessary infrastructure [R96,WNE00]. As in other network models, a challenging task is to enable fast and *reliable* communication.

A wireless network can be modeled as a directed graph G where an edge (u, v) exists if and only if u can communicate with v in one hop. Communication

between two stations that are not adjacent can be achieved by *multi-hop* transmissions. A useful (and sometimes unavoidable) paradigm of wireless communication is the structuring of communication into synchronous *time-slots*. This paradigm is commonly adopted in the practical design of protocols and hence its use in theoretical analysis is well motivated [BG187,CGGPR00,G85,R72]. In every time-slot, each active node may perform local computations and either transmit a message along all of its outgoing edges or try to recover messages from all its incoming edges (the last two operations are carried out by means of an omnidirectional antenna). This feature is extremely attractive in its *broadcast* nature: a single transmission by a node could be received by all its neighbors within one time-slot. However, we adopt the model (see [ABLP89,BG187,CGGPR00]) in which a single radio frequency is used: when two or more neighbors of a node are transmitting at the same time-slot, a *collision* occurs and the message is lost. So, a node can recover a message from one of its incoming edges if and only if this edge is the only one bringing in a message.

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One of the fundamental tasks in wireless network communication is the *broadcast* operation. It consists in transmitting a message from one source node to all the nodes. Most of the proposed broadcast protocols in wireless networks concern the case in which the network is *fault free*. However, wireless networks are typically adopted in scenarios where unpredictable node and link faults happen very frequently.

Node failures happen when some hardware or software component of a station does not work, while link failures are due to the presence of a new (artificial or natural) hurdle that does not allow the communication along that link. Typically, while it is reasonable to assume that nodes know the initial topology of the network, they know nothing about the duration and the location (in the network) of the faults. Such faults may clearly happen at any instant, *even during the execution of a protocol*. In the sequel, such kind of faults will be called *dynamical faults* or, simply, faults.

The (worst-case) completion-time of a fault-tolerant broadcasting protocol on a graph G is defined as the maximum number (over all possible fault patterns) of time-slots required to *inform* all nodes in the fault-free part of the network which are reachable from the source (a more formal definition will be given in Section 2). The aim of this paper is thus to investigate the completion time of such broadcast protocols in presence of *dynamical faults*.

1.1. Previous results

(*Fault-free*) *broadcasting*: We will mention only the best results which are presently known for the fault-free model. An $O(D + \log^5 n)$ upper bound on the completion time for n -node networks of *source eccentricity* D is proved in [GM95]. The source eccentricity is the maximum oriented distance (i.e. number of hops) from the source s to a reachable node of the network. Notice that D is a trivial lower bound for the broadcast operation. In [CCMPS01], the authors give a protocol that completes broadcasting within $O(D \log \Delta \log(n/D))$ time, where Δ denotes the maximal in-degree of the network. This bound cannot be improved in general: [ABLP89] provides an $\Omega(\log^2 n)$ lower bound that holds for graphs of maximal eccentricity $D = 2$.

In [CK85], the authors show that scheduling an optimal broadcast is NP-hard. The APX-hardness of the problem is proved in [CCMPS01].

Permanent-fault-tolerant broadcasting: A node has a *permanent* fault if it never sends or receives messages since the beginning of the execution of the protocol. In [KKP98], the authors consider the broadcasting operation in presence of permanent unknown node faults for two restricted classes of networks: linear and square (or

hexagonal) meshes. They consider both *oblivious* and *adaptive* protocols: in the former case, all transmissions are scheduled in advance; in particular, the action of a node in a given time-slot is independent of the messages received so far. In the latter case, nodes can decide their action also depending on the messages received so far. For both the cases, the authors assume the existence of a bound t on the number of faults and, then, they derive a $\Theta(D + t)$ bound for oblivious protocols and a $\Theta(D + \log \min\{\Delta, t\})$ bound for adaptive protocols, where, in this case (and in the sequel), D denotes the source eccentricity in the fault-free part of the network, i.e., the *residual* eccentricity.

More recently, the issue of permanent-fault-tolerant broadcasting on general networks has been studied in [CGGPR00, CGOR00, CGR00, CMS01]. Indeed, in these papers, several lower and upper bounds on the completion time of broadcasting are obtained on the *unknown* fault-free network model. A wireless network is said to be *unknown* when every node knows nothing about the network but its own label. Even though it has never been observed, it is easy to show that a broadcasting protocol for unknown fault-free networks is also a permanent-fault-tolerant protocol for general (known) networks and vice versa. So, the results obtained in the unknown model immediately apply to the permanent-fault tolerance issue. In particular, one of the results in [CMS01] can be interpreted as showing the existence of an infinite family of networks for which any permanent-fault-tolerant protocol is forced to perform $\Omega(n \log D)$ time-slots to complete broadcast. The best general upper bound for permanent-fault-tolerant protocols is $O(n \log^2 n)$ [CGR00]. This protocol is thus almost optimal when $D = \Omega(n^\alpha)$ for any constant $\alpha > 0$. In [CMS01], the authors provide a permanent-fault-tolerant protocol having $O(D \Delta \log^2 n)$ completion time on any network of maximum in-degree Δ .

Other models: A different kind of fault-tolerant broadcasting is studied in [PR97]: they in fact introduce fault-tolerant protocols that work under the assumption that all faults are *eventually* repaired. The protocols are not analyzed from the point of view of worst-case completion time. Finally, in [KM98], the case in which broadcasting messages may be corrupted with some probability distribution is studied.

Dynamical-fault-tolerant broadcasting: We observe that permanent faults are special cases of dynamical faults and, moreover, we emphasize that *all* the above protocols *do not work* in presence of dynamical faults. This is mainly due to the collisions yielded by any unpredictable wake-up of a faulty node/link during the protocol execution.

To the best of our knowledge, broadcasting on wireless networks in presence of dynamical faults has never been studied before.

1.2. Our results

Oblivious protocols: A simple oblivious (*dynamical*) *fault-tolerant distributed broadcasting* (FDB) protocol relies on the *Round Robin* scheduling: given an n -node network G and a source node s , the protocol runs a sequence of consecutive identical *phases*; each phase consists of n time-slots and, during the i th time-slot ($i = 1, \dots, n$), node i , if *informed*,¹ acts as transmitter while all the other nodes work as receivers. It is not hard to show that, for any fault pattern yielding residual source eccentricity D , the Round Robin protocol completes broadcasting on G , after D phases (so, after Dn time-slots). One may think that this simple oblivious FDB protocol is not efficient (or, at least, not optimal) since it does *never* exploit simultaneous transmissions.

Rather surprisingly, we show that, for any n and for any $D < n$, it is possible to define an n -node network G , such that *any* oblivious FDB protocol requires $\Omega(Dn)$ time-slots to complete broadcast on G . It thus follows that the Round Robin protocol is *optimal* on general networks. The proof departs significantly from the techniques used in all previous related works (such as those used for the case of permanent faults and based on *selective families* [CGGPR00,CMS01]): it in fact relies on a tight lower bound on the length of D -sequences (see Definition 3.1), a combinatorial tool that may have a per se combinatorial interest.

We then show that a broad class of wireless networks admit an oblivious FDB protocol which is faster than the Round Robin protocol. Indeed, we exploit small *ad hoc strongly selective families*, a variant of strongly selective families (also known as *superimposed codes* [CMS01,I97]), in order to develop an oblivious FDB protocol that completes broadcasting within $O(D \min\{n, \Delta \log n\})$ time-slots, where Δ is the maximum in-degree of the input network. This protocol is thus faster than the Round Robin protocol for all networks such that $\Delta = o(n/\log n)$ and it is almost optimal for constant Δ .

Adaptive protocols: In adaptive FDB protocols, nodes have the ability to decide their own scheduling as a function of the messages received so far. A natural and interesting question is whether adaptive FDB protocols are faster than oblivious ones. We give a partially negative answer to this question and show that, when $D = \Theta(\sqrt{n})$, any adaptive FDB protocol requires $\Omega(Dn)$ time-slots. This result is obtained by strengthening the connection between strong selectivity and the task of fault-tolerant broadcasting: we indeed exploit the tight lower bound on the size of strongly selective families (see Definition 3.2), given in [CMS01]. This implies that *no*

(adaptive) FDB protocol can achieve $o(Dn)$ completion time on arbitrary networks.

2. Preliminaries

The aim of this section is to formalize the concept of FDB protocol and its completion time.

According to the fault-tolerance model adopted in the literature [KKP98,P02], an FDB protocol for a graph G is a broadcasting protocol that, for any source s , and for any (node/link) fault pattern F , guarantees that every node, which is reachable from s in the *residual subgraph* G^F , will receive the source message. A fault pattern F is a function that maps every time-slot t to the subset $F(t)$ of nodes and links that are faulty at time-slot t . The residual subgraph G^F is the graph obtained from G by removing all those nodes and links that belong to $F(t)$, for some time-slot t during the execution of the protocol. The completion time of the protocol on a graph G and source s is the maximal (over all possible fault patterns) number of time-slots to perform the above task.

This definition implies that nodes that are not reachable from the source in the residual subgraph are not considered in the analysis of the completion time of FDB protocols. We emphasize that any attempt to consider a larger residual subgraph makes the worst-case completion time of *any* FDB protocol unbounded.

3. Oblivious fault-tolerant protocols

3.1. The lower bound

An oblivious protocol for an n -node network can be represented as a sequence $S = (S_1, S_2, \dots, S_l)$ of transmissions, where $S_i \subseteq [n]$ is the set of nodes that transmit during the i th time-slot and l denotes the worst-case (w.r.t. all possible fault patterns) completion time. Since the protocols are oblivious, we also assume that, if a node belongs to S_i (so it should transmit at time-slot i) but it has not received the source message during the first $(i - 1)$ time-slots, then it will send no message at time-slot i .

In order to prove the lower bound, we consider the complete directed graphs K_n , $n \geq 1$. We first show that any oblivious FDB protocol S on K_n must satisfy the following property.

Definition 3.1. A sequence $S = (S_1, S_2, \dots, S_l)$ of subsets of $[n]$ is called a D -sequence for $[n]$ if, for each subset H of $[n]$ with $|H| \leq D$ and each permutation $\pi = (\pi_1, \pi_2, \dots, \pi_{|H|})$ of H , there exists a subsequence

¹ A node is *informed* during a time-slot t if it has received the source message in some time-slot $t' < t$.

$(S_{i_1}, S_{i_2}, \dots, S_{i_{|H|-1}})$ of $\mathbf{S}$ such that

$$\pi_j \in S_{i_j} \quad \text{and} \quad S_{i_j} \subseteq H \quad \text{for } 1 \leq j < |H|.$$

Lemma 3.1. *For any $n > 0$ and $D < n$, let $\mathbf{S}$ be an oblivious protocol which completes broadcast on the graph K_n for every fault pattern yielding a residual subgraph of source eccentricity at most D . Then $\mathbf{S}$ is a D -sequence for $[n]$.*

Proof. Let $\mathbf{S} = (S_1, S_2, \dots, S_l)$ be an oblivious FDB protocol which completes broadcast on K_n for every fault pattern yielding a residual subgraph of source eccentricity at most D . Let us consider a subset H of $[n]$ with $|H| \leq D$ and a permutation $\pi = (\pi_1, \pi_2, \dots, \pi_{|H|})$ of H . We will define a fault pattern F of K_n and a source node π_0 such that

1. π_0 has residual eccentricity $|H|$,
2. the fact that $\mathbf{S}$ completes broadcast for the pattern F implies the existence of a subsequence $(S_{i_1}, \dots, S_{i_{|H|-1}})$ of $\mathbf{S}$ such that $\pi_j \in S_{i_j}$ and $S_{i_j} \subseteq H$, for $1 \leq j < |H|$.

Let us choose the source π_0 as a node in $[n] \setminus H$ and consider the set of (directed) edges

$$A = \bigcup_{0 \leq i < |H|} (\pi_i, \pi_{i+1}).$$

The pattern F is defined as follows:

for any $i \geq 1$ and for any $u \in [n]$ with $u \neq \pi_{i-1}$,
the edge (u, π_i) is faulty at time slot t if and
only if $\pi_{i-1} \notin S_t$ or π_{i-1} is not yet informed
at that time slot. In other words, the edge
 (u, π_i) is faulty whenever π_{i-1} cannot inform π_i .

Observe that the edges in A are never faulty. Moreover, it is easy to verify that the residual source eccentricity is $|H|$. Since, by definition, the protocol $\mathbf{S}$ completes the broadcast on any residual subgraph of source eccentricity at most D , then the protocol $\mathbf{S}$ completes the broadcast on the graph K_n^F . So, for any $i > 0$, there is a time-slot in which π_{i+1} gets informed. By definition of F , the only node that can inform π_{i+1} is π_i . Since all nodes in $[n] \setminus H$ are informed during the first time-slot in which the source π_0 transmits, then π_i can inform π_{i+1} at a time-slot t only if $S_t \subseteq \{\pi_1, \pi_2, \dots, \pi_{|H|}\}$. Thus, there must exist a subsequence $(S_{i_1}, S_{i_2}, \dots, S_{i_{|H|-1}})$ of $\mathbf{S}$ such that $\pi_j \in S_{i_j}$ and $S_{i_j} \subseteq H$, for $1 \leq j < |H|$. $\square$

We now prove a tight lower bound on the length of a D -sequence. To this aim, we need the following technical result.

Lemma 3.2. *If $\mathbf{S}$ is a D -sequence for $[n]$, then*

$$\sum_{S \in \mathbf{S}} |S| = \Omega(Dn).$$

Proof. Let $\mathbf{S} = (S_1, S_2, \dots, S_l)$ be a D -sequence for $[n]$ and consider the sequence $(k_1, k_2, \dots, k_{D-1})$ defined (by induction) as follows:

$$k_1 = \min \left\{ h \mid [n] = \bigcup_{1 \leq j \leq h} S_j \right\}.$$

By definition of D -sequence, k_1 must exist and so there exists (at least one) element in the set

$$S_{k_1} \setminus \bigcup_{1 \leq j < k_1} S_j.$$

Then, let π_1 be any of such elements. We now assume that the indices $k_1, k_2, \dots, k_i$ and the elements $\pi_1, \pi_2, \dots, \pi_i$ are already defined, then

$$k_{i+1} = \min \left\{ h \mid ([n] - \{\pi_1, \dots, \pi_i\}) \subseteq \bigcup_{k_i < j \leq h} S_j \right\}$$

(again, we notice that k_{i+1} must exist since $\mathbf{S}$ is a D -sequence and $i+1 < D$) and let π_{i+1} be any element in

$$([n] - \{\pi_1, \dots, \pi_i\}) \cap \left(S_{k_{i+1}} \setminus \bigcup_{k_i < j < k_{i+1}} S_j \right).$$

By definition of the above sequence, it holds that

$$\sum_{j=1}^{k_1} |S_j| = n \quad \text{and} \quad \sum_{k_i < j \leq k_{i+1}} |S_j| \geq n - i$$

for any $i = 1, \dots, D-1$.

It thus follows that

$$\begin{aligned} \sum_{S \in \mathbf{S}} |S| &\geq \sum_{j=1}^{k_1} |S_j| + \sum_{i=1}^{D-2} \sum_{k_i < j \leq k_{i+1}} |S_j| \geq n + \sum_{i=1}^{D-2} (n - i) \\ &= \Omega(Dn). \quad \square \end{aligned}$$

Lemma 3.3. *If $\mathbf{S}$ is a D -sequence for $[n]$ then $|\mathbf{S}| = \Omega(Dn)$.*

Proof. We count in two different ways the following number:

$$N = |\{(H, (S, x)) \mid H \subseteq [n], S \in \mathbf{S}, S \subseteq H \text{ and } x \in S\}|.$$

Let us consider H and the subsequence $\mathbf{S}_H$ of $\mathbf{S}$ obtained by deleting from $\mathbf{S}$ all the sets S such that $S \not\subseteq H$. By definition of D -sequence $\mathbf{S}_H$ must be a D -sequence for H . Hence, from Lemma 3.2, we have at least $cD|H|$ ways of choosing (S, x) , for a constant $c > 0$. Thus

$$\begin{aligned} N &\geq \sum_{H \subseteq [n]} cD|H| = cD \sum_{i=1}^n \binom{n}{i} i > cD \sum_{i=\lceil \frac{n}{2} \rceil}^n \binom{n}{i} i \\ &> \frac{cDn}{4} 2^n. \end{aligned} \tag{1}$$

Now let (S, x) be fixed. There are $2^{n-|S|}$ subsets H of $[n]$ such that $S \subseteq H$. Thus

$$N = \sum_{S \in \mathcal{S}} |S| \cdot 2^{n-|S|} = 2^n \sum_{S \in \mathcal{S}} \frac{|S|}{2^{|S|}} \leq 2^n \sum_{S \in \mathcal{S}} \frac{1}{2} = \frac{2^n}{2} |\mathcal{S}|. \quad (2)$$

Finally, by comparing Eqs. (1) and (2), we derive

$$\frac{2^n}{2} |S| > \frac{cDn}{4} 2^n,$$

so

$$|S| > \frac{cDn}{2}. \quad \square$$

We are now able to show the following.

Theorem 3.1. *Let $n > 0$ and $1 \leq D \leq n - 1$. For every oblivious FDB protocol on the graph K_n , there exist a source s and a fault pattern F such that*

- (i) *the residual graph has eccentricity not larger than D ,*
- (ii) *the protocol is forced to perform $\Omega(Dn)$ time-slots.*

Proof. Let $\mathbf{S}$ be any oblivious FDB protocol for K_n . Define T as the maximum completion-time of $\mathbf{S}$ over all possible sources and all possible fault patterns yielding a residual subgraphs of source eccentricity at most D . Observe that the first T transmission sets of $\mathbf{S}$ must constitute a protocol which completes broadcast on K_n for every fault pattern yielding a residual source eccentricity at most D . Hence, from Lemma 3.1, the first T transmission sets of $\mathbf{S}$ must be a D -sequence for $[n]$. From Lemma 3.3, it must hold that $T = \Omega(Dn)$. $\square$

The above theorem implies that a bound on the completion time which is independent from D cannot be asymptotically “smaller” than n^2 .

Corollary 3.1. *For any $n > 0$, any oblivious FDB protocol completes broadcasting on the graph K_n in $\Omega(n^2)$ time-slots.*

3.2. Efficient protocols for networks of “small” in-degree

In this subsection, we show that networks of maximum in-degree $\Delta = o(n/\log n)$ admit oblivious FDB protocols which are faster than the Round Robin one. To this aim, we need the following combinatorial tool.

Definition 3.2. Let $\mathcal{S}$ and $\mathcal{N}$ be families of sets. The family $\mathcal{S}$ is strongly selective for $\mathcal{N}$ if for every set

$N \in \mathcal{N}$ and, for every element $x \in N$, there exists a set $S \in \mathcal{S}$ such that $N \cap S = \{x\}$.

Strongly selective families for the family of all subsets of $[n]$ having size at most Δ have been recently used to develop multi-broadcast protocols on the unknown model [CMS01, CMSb01]. The following protocol instead uses strong selectivity for the family $\mathcal{N}$ consisting of the sets of in-neighbors of the nodes of the input network G . Indeed, for each node v of G , let $N(v) \subseteq [n]$ be the set of its in-neighbors and let $\mathcal{N} = \{N(v) \mid v \in V\}$. Let $\mathcal{S} = \{S_1, S_2, \dots, S_m\}$ be any (arbitrarily ordered) strongly selective family for $\mathcal{N}$.

Description of Protocol SELECT. The protocol consists of a sequence of phases:

- In the first phase the source sends its message.
- All successive phases are identical and each of them consists of m time-slots. At time-slot j of every phase, any informed node v sends the source message if and only if it belongs to S_j ; All the remaining nodes act as receivers.

Lemma 3.4. *For any (dynamical) fault pattern F , at the end of phase i , every node at distance i , from the source s in the residual subgraph G^F , is informed. So, SELECT completes broadcasting within Dm time-slots, where D is the residual source eccentricity.*

Proof. The proof is by induction on the distance i . For $i = 1$ it is obvious. We thus assume that all nodes at distance i have received the source message during the first i phases. Consider a node v at distance $i + 1$ in the residual subgraph and a node $u \in N(v)$ at distance i in the residual subgraph. Notice that, since v is at “residual” distance $i + 1$ from s , such an u must exist. Moreover, $N(v)$ belongs to $\mathcal{N}$ and $\mathcal{S}$ is strongly selective for $\mathcal{N}$, so there will be a time-slot in phase $i + 1$ in which only u (among the nodes in $N(v)$) transmits the source message and v will successfully receive it.

It is now clear that the total number of time-slots required by the protocol to complete the broadcast is Dm . $\square$

The above lemma motivates our interest in finding strongly selective families of small size since the latter is a factor of the completion time of our protocol.

Lemma 3.5. *For any family $\mathcal{N}$ of sets, each of size at most Δ , there exists a strongly selective family $\mathcal{S}$ for $\mathcal{N}$ such that $|\mathcal{S}| = O(\Delta \max\{\log |\mathcal{N}|, \log \Delta\})$.*

Proof. We assume, without loss of generality, that the sets in $\mathcal{N}$ are subsets of the ground set $[n]$ and that $\Delta \geq 2$

(for $\Delta = 1$ the family $\mathcal{S} = \{[n]\}$ trivially proves the lemma).

We use a probabilistic argument: construct a set S by picking every element of $[n]$ with probability $\frac{1}{\Delta}$. For fixed $N \in \mathcal{N}$ and $x \in N$ it holds that

$$\Pr[N \cap S = \{x\}] = \frac{1}{\Delta} \left(1 - \frac{1}{\Delta}\right)^{|N|-1} \geq \frac{1}{\Delta} \left(1 - \frac{1}{\Delta}\right)^{\Delta} \geq \frac{1}{4\Delta} \quad (3)$$

(where the last inequality holds since $\Delta \geq 2$). Consider now a family $\mathcal{S} = \{S_1, S_2, \dots, S_m\}$ where each set S_i is constructed independently as above. From inequality (3), it follows that the probability that $\mathcal{S}$ is not strongly selective for fixed $N \in \mathcal{N}$ and $x \in N$ is at most

$$\left(1 - \frac{1}{4\Delta}\right)^m \leq e^{-\frac{m}{4\Delta}}$$

(the above bound follows from the well-known inequality $1 - t \leq e^{-t}$ that holds for any real t). It thus follows that

$$\Pr[\mathcal{S} \text{ is not strongly selective for } \mathcal{N}] \leq \sum_{N \in \mathcal{N}} \sum_{x \in N} e^{-\frac{m}{4\Delta}} \leq |\mathcal{N}| \Delta e^{-\frac{m}{4\Delta}}$$

and the last value is less than 1 for $m > 8\Delta \max(\log |\mathcal{N}|, \log \Delta)$. $\square$

We emphasize that the probabilistic construction of strongly selective families in the above proof can be efficiently (i.e. in polynomial time) de-randomized by means of a suitable application of the method of *conditional probabilities* [MR95]. This technique has been recently applied to a weaker version of selectivity in [CCMPS01]. Furthermore, as for strong selectivity, we can also use the deterministic efficient construction of *superimposed codes* given in [I97] that yields strongly selective families of (larger) size $O(\Delta \log(|\mathcal{N}|) \log^2 n)$.

Theorem 3.2. *The oblivious FDB protocol SELECT completes broadcast within $O(D \min\{\Delta \log n, n\})$ time-slots on any n -node graph G with maximum in-degree Δ , where D is the residual source eccentricity.*

Proof. Since the graph has maximum in-degree Δ , the size of any subset in $\mathcal{N}$ is at most Δ . Hence, from Lemma 3.5, there exists a strongly selective family $\mathcal{S}$ for $\mathcal{N}$ of size $|\mathcal{S}| \leq \min\{c\Delta \log n, n\}$, for some constant $c > 0$ (the bound n is due to the fact that a family of n singletons is always strongly selective). The theorem is thus an easy consequence of the above bound and Lemma 3.4. $\square$

4. Adaptive fault-tolerant protocols

In this section, a lower bound on the completion-time of adaptive FDB protocols is given. To this aim, we consider strongly selective family for the family of all subsets of $[n]$ having size at most Δ (in short, (n, Δ) -strongly selective families). As for the size of such families, the following lower bound is known.

Theorem 4.1 (Clementi et al. [CMS01]). *If $\mathcal{S}$ is an (n, Δ) -strongly selective family then it holds that*

$$|\mathcal{S}| = \Omega\left(\min\left\{\frac{\Delta^2}{\log \Delta} \log n, n\right\}\right).$$

We will adopt the general definition of (adaptive) distributed broadcast protocols introduced in [BG187]. In such protocols, the action of a node in a specific time-slot is a function of its own label, the number of the current time-slot t , the input graph G , and the messages received during the previous time-slots.

Theorem 4.2. *Let $n > 0$. For every FDB protocol P , there exist a source s and a fault pattern for the graph K_n such that the residual graph has eccentricity $\Theta(\sqrt{n})$ and P completes broadcasting on K_n in $\Omega(n\sqrt{n})$ time-slots.*

Proof. Given any protocol P , we define two sets of faults: a set of edges of K_n that suffer a permanent fault (i.e., they will be forever down since the beginning of the protocol) and a set of dynamical node faults. The permanent faults are chosen in order to yield a layered graph G^P which consists of $D + 1$ levels $L_0, L_1, \dots, L_D$ where $D = \lfloor \sqrt{n}/2 \rfloor$. Level L_0 contains only the source s , level L_j , $j < D$, consists of at most Δ nodes with $\Delta = \sqrt{n}$ and L_D consists of all the remaining nodes. All nodes of L_{j-1} in G^P have (only) outgoing edges to all nodes in L_j .

Both permanent and dynamical faults will be determined depending on the actions of P . As for permanent faults, instead of describing the set of faulty edges, we provide (in the proof of Claim 1), for any $j = 1, \dots, D$, the set of nodes that belongs to L_j . This permanent fault pattern will be combined with the dynamical fault pattern (which is described below) in such a way that the protocol is forced to execute $\Omega\left(\frac{\Delta^2}{\log \Delta} \log n\right) = \Omega(n)$ time-slots in order to successfully transmit the initial message between two consecutive levels.

From Theorem 4.1, there exists a constant $c > 0$ such that, any $(\lceil n/2 \rceil, \Delta)$ -strongly selective family must have size at least T , where $T \geq cn$. The theorem is thus an easy consequence of the following.

Claim 1. For any $0 \leq j \leq D$, there exists a node assignment to levels $L_0, \dots, L_j$ and a pattern of dynamical node faults in these levels such that

- (a) The residual graph contains (at least) a path from s to a node in L_j ,
- (b) P does not broadcast the source message to any node in $V \setminus \bigcup_{i=1}^j L_i$ before the time-slot $j \cdot T$.

Proof. The proof is by induction on j . For $j = 0$, $L_0 = \{s\}$ and the claim is trivial. We thus assume the thesis be true for $j - 1$. Let us define

$$R = \{v \mid v \text{ is not already assigned to levels } L_0, \dots, L_{j-1}\}.$$

Since, for any $i \geq 1$, $|L_i| \leq \sqrt{n}$ and $j < D = \lceil \sqrt{n}/2 \rceil$, then $|R| \geq \lceil n/2 \rceil$. Let L be an arbitrary subset of R . Consider the following two cases: (i) L_j is chosen as L ; (ii) L_j is chosen as R (i.e., all the remaining nodes are assigned to the $(j+1)$ th level). In both cases, the predecessor² subgraph G_u^P of any node $u \in L$ is that induced by $L_0 \cup L_1 \cup \dots \cup L_{j-1} \cup \{u\}$ in G^P . It follows that the behavior of node u , according to protocol P , is the same in both cases. We can thus consider the behavior of P when $L_j = R$. Then, we define

$$S_t = \{u \in R \mid u \text{ acts as transmitter at time-slot } (j-1)T + t\}$$

and the family $\mathcal{S} = \{S_1, \dots, S_{T-1}\}$ of subsets from R . Since $|\mathcal{S}| < T$, by definition of T , $\mathcal{S}$ is not $(\lceil n/2 \rceil, \Delta)$ -strongly selective; so, a subset $L \subset R$ exists such that $|L| \leq \Delta$ and L is not strongly selected by $\mathcal{S}$ (and thus by P) in any time-slot t such that $(j-1)T + 1 \leq t \leq jT - 1$. Then, we choose L_j as L and all the edges from L_{j-1} to L_j are never faulty. Let thus u be a node in L which is not selected. At every time-slot $t \in [(j-1)T + 1, \dots, jT - 1]$, if there is exactly one node v in L_j which is scheduled as transmitter then we let v suffer a failure. Otherwise, no fault happens. Thanks to this fault pattern, v never transmits alone before time-slot jT (notice that v never suffers faults and property (a) of the claim holds).

By inductive hypothesis, nodes in L_{j-1} cannot transmit the source message during the first $(j-1)T$ time-slots. Furthermore, thanks to the above fault pattern, the same holds for the time-slots $(j-1)T + 1, \dots, jT - 1$. Property (b) of the Claim is thus proved. $\square$

Corollary 4.1. No FDB protocol can achieve an $o(Dn)$ completion time on general n -node networks.

² Given a graph G , the predecessor subgraph G_u of a node u is the subgraph of G induced by all nodes v for which there exists a directed path from v to u .

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